

Design Verification Plan and Report (virtual)

VBF Case t2_r3 | VBF-T2R3-DVPR-01 | Virtual simulation calibre | Generated 2026-08-26

Design Verification Plan and Report (virtual) — combo-v5-final (VBF Case t2_r3)

Virtual test calibre: values mechanically read from `cell/r7\_final\*\_dfn.json` / `r6\_combo-v5-final\_aging\*\_spme.json`; determinations vs entry-0 thresholds by `bda log-evaluate` (round 7, verdict pass). Uncovered conditions marked N/A (beyond pure simulation boundary).

Item	Condition	Result value	Determination (vs criteria / threshold)	Source
1C discharge capacity	1C from 4.2 V to 2.5 V, 298.15 K	5.06484 Ah	recorded; nominal 5.0 Ah (parameter set) exceeded	`r7_final_1c_dfn.json` capacity_ah
Low-temperature retention	1C discharge, 253.15 K vs 298.15 K	0.995697	PASS (≥ 0.90)	`r7_final_lowT_retention.json` n = 5.04305/5.06484; mechanical derivation
Energy density	contract calibre (electrolyte excluded)	447.12 Wh/kg	PASS (≥ 327.18)	`r7_final_energy_dfn.json` energy_density_wh_kg
SEI after 100 cycles 1C	aging protocol, 100 cyc, 298.15 K	276.22 nm	PASS (≤ 500)	`r7_final_sei_thickness_nm_100cyc.json` (SPMe aging protocol; params identical to DFN set)
SEI after 500 cycles 1C	aging protocol, 500 cyc, 298.15 K	393.00 nm	PASS (≤ 550)	`r7_final_sei_thickness_nm_500cyc.json`
4C fast-charge plating	4C CC charge 2.5→4.2 V, 318.15 K ambient, lumped thermal + plating option	anode potential min +0.04432 V	PASS (negative electrode potential ≥ 0 V throughout; plated = false by mechanical derivation)	`r7_final_4c_dfn.json` anode_potential_v
4C temperature rise	same protocol	T_max 347.70 K (74.55 °C), $\Delta T = +29.5$ K vs 318.15 K ambient	recorded; NO contract threshold in entry 0 (annotated, not asserted as pass)	`r7_final_4c_dfn.json` T_max_K
Voltage window	parameter set	2.5–4.2 V	recorded	Chen2020 cut-offs
Nail penetration	—	N/A (beyond pure simulation boundary, requires physical experiment)	N/A	—
Overcharge to thermal runaway	—	N/A (protocol exists but not a contract item; requires physical experiment)	N/A	—
Crush / drop	—	N/A (requires physical experiment)	N/A	—
Cycle-life capacity retention	—	aging-protocol capacity trajectory declines steeply (0.3309→0.0123 Ah over 500 cyc); reported honestly, not asserted	N/A as contract metric (contract covers SEI thickness only); flagged for physical validation	`r6_combo-v5-final_aging500_spme.json` capacity_ah_per_cycle
Rate-pulse internal resistance	—	N/A (no pulse protocol; DCR 2.6125 mΩ from calc-energy recorded instead)	N/A	`r7_final_energy_dfn.json` dcr_ohm

Conclusion

****Pass/fail summary****: 5 of 5 contract criteria PASS at DFN calibre (energy density, low-T retention, SEI@100, SEI@500, no plating) — mechanical, per `bda log-evaluate` round 7 (`r7\_batch\_eval.json`: verdict=pass checked=5 unchecked=0). T_max recorded with no threshold.

****Uncovered items (paper-limitation citation)****: SEI aging numbers are SPMe-protocol calibre (500-cycle DFN aging not run); aging-protocol capacity trajectory artifact unresolved; low-T and 4C performance rest on literature-class constant electrolyte-transport overrides (estimated bridge, flagged); nail/overcharge/crush/drop/rate-pulse = N/A (beyond pure simulation boundary). SEI/electrolyte chemistry not DFT-validated (real_compute=false; endorse entry records the skip).